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Adjustable firearm steady rest

Abstract

An adjustable shooting steady rest configured to clamp to a wide variety of structures. The steady rest includes a chassis with one or more clamps opposing a clamping bracket. The user places the chassis over a structure such as a safety bar or windowsill. The user then tightens the clamp(s) to lock the chassis in place. The chassis includes a rod receiver. A rod is adjustably attached to the rod receiver. A firearm support is provided on the upper portion of the rod.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. patent application Ser. No. 17/687,965. The parent application was filed Mar. 7, 2022. It listed the same inventor.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT
(1) Not Applicable

(2) Not Applicable

BACKGROUND OF THE INVENTION

1. Field of the Invention

(3) This invention relates to the field of shooting. More specifically, the invention comprises an adjustable firearm support that can be attached to a variety of structures.

2. Description of the Related Art

(4) Those familiar with the shooting sports know that the use of a steady rest greatly increases firearm accuracy. FIGS. **1** and **2** illustrate scenarios in which a steady rest would be beneficial. FIG. **1** illustrates a prior art tree stand **10**. There are many different types of tree stand. Most, however, include a seat assembly **12**, a foot plate **14**, and a safety bar **16**. The example shown in FIG. **1** includes a ladder. Hunter **18** climbs the ladder and sits in seat assembly **12**. Safety bar is in a raised position when the hunter enters or exits the tree stand.

(5) Once seated, the hunter pivots safety bar **16** down to the position shown. The safety bar is pivotally attached to the stand by a pair of pivot joints **15**—only one of which is visible in FIG. **1**. The safety bar is held in place by gravity. It actually provides a suitably stable rest point for shooting, but its position is too low. The hunter in FIG. **1** has assumed a typical shooting stance required to aim and fire rifle **20** from a tree stand. Trigger hand **22** is closed around the grip just to the rear of the trigger guard. Support hand **24** rests beneath the rifle's forend. This is in effect a modified “offhand” shooting position. It cannot be considered a seated shooting position because the hunter's elbows cannot be rested on any portion of the hunter's legs. Offhand is an unstabilized shooting position that significantly limits accuracy.

(6) FIG. **2** shows a second common scenario. Shooting blind **26** conceals hunter **18**. Opening **28** is provided for viewing and shooting. When a shot is to be taken, hunter **18** extends rifle **20** through opening **28**. Sill **30** can provide a steady rest. A towel or sandbag is sometimes placed on the sill or on a shelf provided on the inside of the blind just below the sill. These devices do provide stability but are often cumbersome and inflexible to use. The result is that the hunter may wind up shooting off-hand as depicted in FIG. **2**.

(7) Many other shooting scenarios do of course exist, and the invention is by no means limited to the particular applications depicted. However, in general, it is desirable to provide a portable steady rest that can be clamped to a variety of structures and adjusted in height. The present invention provides such a device.

BRIEF SUMMARY OF THE INVENTION

(8) The present invention comprises an adjustable shooting steady rest configured to clamp to a wide variety of structures. The steady rest includes a chassis with one or more clamps opposing a clamping bracket. The user places the chassis over a structure such as a safety bar or windowsill. The user then tightens the clamp(s) to lock the chassis in place. The chassis includes a rod receiver. A rod is adjustably attached to the rod receiver. A firearm support is provided on the upper portion of the rod. The firearm support is configured to support a suitable portion of a firearm—such as a rifle forend or barrel. The user adjusts the rod's position to provide a desired height for the firearm support.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) FIG. **1** is a perspective view, showing a hunter in a prior art tree stand.

(2) FIG. **2** is a perspective view, showing a hunter in a prior art shooting blind.

(3) FIG. **3** is a perspective view, showing a first embodiment of the inventive steady rest.

(4) FIG. **4** is a side elevation view, showing the embodiment of FIG. **3** clamped to a large tube.

- (5) FIG. 5 is a side elevation view, showing the embodiment of FIG. 3 clamped to a small tube.
- (6) FIG. 6 is a side elevation view, showing the embodiment of FIG. 3 clamped to a square tube.
- (7) FIG. 7 is a side elevation view, showing the embodiment of FIG. 3 clamped to a sill.
- (8) FIG. 8 is a perspective view, showing the invention clamped to a safety bar on a tree stand.
- (9) FIG. 9 is a perspective view, showing the invention clamped to a sill in a shooting blind.
- (10) FIG. 10 is a sectional elevation view, showing an embodiment of the rod receiver using threaded inserts.
- (11) FIG. 11 is a sectional elevation view, showing an embodiment of the clamp mount leg using threaded inserts.
- (12) FIG. 12 is a sectional elevation view, showing an alternate embodiment for the adjustable attachment between the rod and the rod receiver.
- (13) FIG. 13 is a perspective view, showing an upward adjustment of the rod in the version of FIG. 12.
- (14) FIG. 14 is a perspective view, showing a downward adjustment of the rod in the version of FIG. 12.
- (15) FIG. 15 is a perspective view, showing an alternate embodiment of the chassis clamped in a horizontal orientation.
- (16) FIG. 16 is a perspective view, showing the embodiment of FIG. 15 clamped in a vertical orientation.
- (17) FIG. 17 is a top view showing the chassis of FIG. 15.
- (18) FIG. 18 is a side view showing the chassis of FIG. 15.
- (19) FIG. 19 is a front view showing the chassis of FIG. 15.
- (20) FIG. 20 is a side section view showing the chassis of FIG. 15.
- (21) FIG. 21 is a side section view showing how the chassis of FIG. 15 can be used in the horizontal orientation and the vertical orientation.

REFERENCE NUMERALS IN THE DRAWINGS

(22) 10 tree stand 12 seat assembly 14 foot plate 15 pivot joint 16 safety bar 18 hunter 20 rifle 21 forend 22 trigger hand 24 support hand 26 shooting blind 28 opening 30 sill 32 steady rest 34 chassis 36 span leg 38 clamp mount leg 40 back leg 42 clamping bracket 43 first leg 44 screw clamp 45 second leg 46 screw clamp 48 insert 50 insert 52 threaded rod 54 rod receiver 56 firearm support 58 foot 60 large tube 62 small tube 64 square tube 66 tip 68 tip 70 weld 72 weld 74 weld 76 weld 79 barrel 80 insert 82 insert 84 female thread 86 female thread 88 tube 90 smooth rod 92 insert 94 insert 96 spring 98 inner slot 100 outer slot 102 hole 104 lock plate 106 hole 108 lock plate 110 release 112 fulcrum 114 shelf 116 unified chassis 118 vertical mount upper rib 120 vertical mount lower rib 122 vertical mount upper hole 124 vertical mount lower hole 126 horizontal mount upper rib 128 horizontal mount lower rib 130 horizontal mount upper hole 131 shelf 132 horizontal mount lower hole 134 vertical mount lock plate slot 136 horizontal mount lock plate slot

DETAILED DESCRIPTION OF THE INVENTION

- (23) The present invention can be physically realized in a wide variety of embodiments. The following description provides details of a few of these. Many more embodiments will occur to those skilled in the art upon reviewing this disclosure. The scope of the invention should therefore be fixed by the claims rather than the descriptions of the disclosed embodiments.
- (24) In general, the invention includes (1) a chassis configured to clamp to a variety of external structures, and (2) an adjustable rod extending from the chassis, with a firearm support being provided on the rod. FIG. 3 shows a first embodiment of steady rest 32. Chassis 34 in this version is a section of C-channel. The three legs of the C-channel are clamp mount leg 38, span leg 36, and bracket leg 40.
- (25) Span leg 36 will ordinarily be placed above the external structure. Bracket leg 40 will normally be placed on the side of the external structure that is opposite to the position of the hunter.

Clamp mount leg **38** will normally be placed on the same side of the external structure as the hunter. The reader should bear in mind, however, that directional terms such as “above” are arbitrary as steady rest **32** can be attached in a variety of orientations.

(26) Clamp mount leg **38** in this example mounts a pair of screw clamps **44**, **46**. Threaded inserts **48**, **50** are pressed into clamp mount leg **38**. Each insert includes a threaded through-hole sized to accommodate the threads of a screw clamp. Clamping bracket **42** is attached to bracket leg **40**. In operation, the screw clamps **44**, **46** squeeze the external structure against clamping bracket **42**—thereby attaching steady rest **32** to the external structure.

(27) A central purpose of the steady rest is the provision of a stable firearm support **56**. In the example of FIG. **3**, firearm support **56** is provided on an end of threaded rod **52**. Threaded rod **52** is adjustably attached to rod receiver **54**. Rod receiver **54**—in turn—is connected to chassis **34**. The rod receiver includes an internal female thread that engages the male thread on threaded rod **52**. The extension of threaded rod **52** above the position of chassis **34** is accomplished by turning threaded rod **52**. For the embodiment shown the threaded rod is given a right-hand thread. Turning the threaded rod in a clockwise direction moves the threaded rod downward and turning the threaded rod in an anti-clockwise direction moves the threaded rod upward.

(28) Clamping bracket **42** allows the chassis to be clamped to a wide variety of external structures. FIGS. **4-7** provide examples of some of these external structures. In the example of FIG. **4**, the invention is being clamped to large tube **60**—the large tube being the external structure in this case. Large tube **60** is part of the safety bar on a tree stand. It has an external diameter of 1.250 inches (31.75 mm).

(29) Clamp bracket **42** is fixedly attached to bracket leg **40** and span leg **36** by welds **72** and **70** respectively. These welds are preferably just small welds made at each end of the C-channel and clamp bracket **42**. The clamp bracket in this example is a simple angle piece having first leg **43** and second leg **45**. The clamp bracket is rotated 45 degrees so that first leg **43** is rotated 45 degrees with respect to bracket leg **40**. The intersection of the two bracket legs **43**, **45** lies on the central axis of screw clamp **44**. When screw clamp **44** is tightened, foot **58** is pressed against large tube **60** and clamp bracket **42** is pulled against large tube **60** as shown. The result is that three points of contact are established around the perimeter of the tube—thereby providing a solid engagement.

(30) FIG. **4** also shown how rod receiver **54** is joined to the C-channel in this embodiment. Two welds **74**, **76** are made between the rod receiver and the C-channel. These are made at the point of tangency for rod receiver **54** (which in this example is a hollow tube).

(31) FIG. **5** shows the same embodiment being clamped to small tube **62**. Small tube **62** has an outside diameter of 0.875 inches (22.22 mm). Clamp bracket **42** accommodates the small tube. Tightening screw clamp **44** again propels foot **58** against the side of the tube—creating three points of contact and a solid engagement. The example shown works well for even smaller tubes—such as one having an outside diameter of only 0.500 inches (12.70 mm).

(32) FIG. **6** shows the same embodiment being clamped to square tube **64**. Square tube **64** has external dimensions of 1.000 inches by 1.000 inches (25.40 mm by 25.40 mm). Clamp bracket **42** accommodates this shape as well—again creating three points of solid contact around the perimeter of the external structure being clamped.

(33) The examples of FIGS. **4-6** apply to tubes commonly used for the safety bar and other structures in tree stands. FIG. **7** pertains to structures such as are present in a blind like the one seen in FIG. **2**. FIG. **7** shows the region of sill **30**. Most blinds have some sort of sill across the bottom of each opening. These are often thicker than the tubes found in tree stands. A typical sill is 1.000 inches to 1.750 inches thick (25.40 mm to 44.45 mm). Span leg **36** is made long enough to accommodate a sill in the clamping gap. In the example shown, two tips **66**, **68** of the legs of clamp bracket **42** bear against the outward facing side of sill **30**. Foot **58** bears against the inward facing side of the sill when screw clamp **44** is tightened.

(34) Many blinds include a shelf **114** just below sill **30**. Such a shelf is useful for holding loose

items like binoculars and ammunition boxes. However, such a shelf can also interfere with the operation of rod **52** if rod receiver **54** is placed on the side of the chassis that customarily resides inside the blind. For this reason, rod receiver **54** is placed on the side of the chassis that customarily resides outside the blind. The screw clamps are placed on the side that customarily resides inside the blind-so they can be more easily accessed by the hunter.

(35) FIGS. **8** and **9** show the invention in use. In FIG. **9**, steady rest **32** has been clamped to safety bar **16** of a tree stand. Threaded rod **52** has been adjusted vertically to place firearm support **56** at a desired height. Forend **21** of the rifle is placed on the firearm support and a steady shooting configuration is thereby created.

(36) FIG. **9** shows the invention in use in a blind. Steady rest **32** has been clamped to sill **30**. Threaded rod **52** has again been adjusted to a desired height for firearm support **56**. In this example, rifle barrel **79** is actually placed in the support **56**. A stable shooting configuration is again thereby created.

(37) FIGS. **10** and **11** illustrate some construction details for the exemplary embodiment of FIG. **3**. It is desirable to minimize the weight of steady rest. To that end, aluminum can be used for the chassis. Aluminum, however, may not provide adequate thread strength for the clamp screws. Inserts can be used to provide adequate thread strength and to minimize the need for potentially expensive threading operations.

(38) FIG. **10** provides a sectional elevation view through the middle of rod receiver **54**. In this version a simple hollow tube **88** provides the structure. Insert **80** is pressed into the open upper end of the tube. It includes female thread **84**. Insert **82** is pressed into the open lower end of the tube. The two inserts include a female threaded passage **84**. The length L of the hollow tube is set so that the threads in insert **82** align with the helical path of the threads in insert **80**. This alignment allows a single threaded rod **52** to thread through insert **80** and insert **82**.

(39) In the preferred embodiments inserts **80**, **82** are clinching steel inserts such as are manufactured by Penn Engineering and Materials, Inc., of Danboro, Pennsylvania, U.S.A., These inserts are assembled by providing (such as by punching or drilling) a properly sized hole and squeezing the insert into the hole (using fixturing on both sides). The insert plastically deforms the perimeter of the hole and becomes clinched to the base material.

(40) FIG. **11** shows another similar insert that is pressed into clamp mount leg **38** in order to engage the threads of a screw clamp. Insert **48** has female thread **86** that is configured to match the thread of the screw clamps. Insert **48** is clinched to the material for clamp mount leg **38**. The engagement is quite strong, so that the insert can withstand the forces generated by the screw clamp and transmit them to the chassis.

(41) The example of FIGS. **3-11** used a threaded engagement to provide the adjustment between rod **52** and rod receiver **52**. This is a very practical and robust adjustable engagement. Other adjustable engagements are possible, however. FIGS. **12-14** illustrate one of these other options.

(42) The embodiment of FIGS. **12-14** uses a modified rod receiver. FIG. **13** is a perspective view of this embodiment. Rod receiver **54** is again attached to chassis **34**—such as by welding. FIG. **12** is a sectional view through the middle of rod receiver **54**. Smooth rod **90** has been substituted for threaded rod **52**. Insert **92** is pressed into the top of tube **88** and insert **94** is pressed into the bottom. The inserts **92**, **94** are preferably clinching steel inserts as described previously. However, each insert **92**, **94** has a smooth bore through its center rather than a threaded bore. Smooth rod **90** is configured to smoothly slide through the two inserts. The bore through the two inserts **92**, **94** in this example is 0.380 inches (9.652 mm). The diameter of the smooth rod in this example is 0.372 inches (9.449 mm). This combination allows guided but smooth motion of smooth rod **90** within inserts **92**, **94**.

(43) A unidirectional latch is incorporated to hold rod **90** in position once a desired elevation has been set. The unilateral latch in this example is similar to the lock plate latch used in the drive mechanism of many caulking guns. Lock plate **108** is provided with a hole that is slightly larger

than the diameter of smooth rod **90**. Stacking lock plate **104** is provided with a hole **102** of the same diameter. In this example, the diameter of holes **102**, **106** is 0.390 inches (9.906 mm). Inner slot **98** is provided in tube **88** proximate mounting leg **40**. Outer slot **100** is provided in the opposite side of the tube.

(44) The ends of the two lock plates **104**, **108** nearest mounting leg **40** rest within inner slot **98**. The lower portion of inner slot **98** becomes fulcrum **112**. Compression spring **96** is sized to slide over smooth rod **90**. This spring is compressed between the lower portion of insert **92** and lock plate **104**. The spring tends to urge the two lock plates **104**, **108** downward. The end of lock plate **108** within inner slot **98** is urged against fulcrum **112** and its further downward motion is thereby arrested. The two lock plates then pivot in an anticlockwise position until further pivoting is prevented by the edges of holes **102**, **106** binding against smooth rod **90**.

(45) In studying the geometry of FIG. **12**, those skilled in the mechanical art will readily appreciate the following: (1) If a user grasps smooth rod **90** and pulls it upward, the rod will slide smoothly upward through the unidirectional latch created by the two lock plates, (2) If a user attempts to push smooth rod **90** downward, the two lock plates will immediately bind against the rod and prevent further downward motion. Thus, rod **90** can move up but cannot move down.

(46) Of course, there will be times when the user wishes to move smooth rod **90** down. Release **110**—which is simply an extension of lock plate **108** is provided for this purpose. If the user urges release **110** upward, then the edges of holes **102**, **106** will disengage from their binding position against smooth rod and the smooth rod will be free to descend. Significantly, however, the user has to keep release **110** in a raised position in order for rod **90** to descend. As soon as the user lets go of release **110**, spring **96** will pivot the lock plates back to the position shown in FIG. **12** and any further downward motion of rod **90** will be prevented.

(47) FIGS. **13** and **14** illustrate the practical application of this unidirectional latch. In FIG. **13**, the user wishes to raise the position of firearm support **56**. The user simply grabs the support—or some portion of smooth rod **90**—and pulls it upward. The support freely moves upward and remains in position when it is released. Thus, the user is able to incrementally adjust firearm support **56** upward until a desired position is reached.

(48) FIG. **14** shows a lowering operation. In order to lower firearm support **56**, the user must pull up on release **110** as indicated by the arrow. Rod **90** is then free to move down. The speed of descent can be regulated by introducing a frictional element. As an example, insert **94** can include a polymer O-ring that bears against smooth rod **90** and slows its descent when the unidirectional latch is released. Returning to FIG. **12**, an exemplary O-ring **116** is provided within insert **94**. In this example, a DELRIN O-ring is used. This O-ring adds friction so that the rod and attached firearm support does not abruptly drop when release **110** is pulled upward. The reader will thereby appreciate that the user can rapidly and easily adjust the position of rifle support **56** using the unidirectional latch and the selective release **110**.

(49) It is preferable to avoid a weldment for the chassis in higher-volume production. FIG. **15** shows an alternate embodiment-denoted as unified chassis **116**. Unified chassis **116** has a constant cross-section. It is preferably made as an extrusion. The details of the extrusion will be described briefly before turning to its functional advantages.

(50) FIG. **18** shows a side elevation view of unified chassis **116**. The reader will note that it includes clamp mounting leg **38**, span leg **36**, and clamping bracket **42** as for the embodiment of FIG. **4**. The function of these elements is the same as for the embodiment of FIG. **4**. Unified chassis **116** contains additional features that allow the use of the unidirectional latch for the firearm support (as in the example of FIGS. **12-14**) without the need for a separate tube **54** to house it. These features also allow the chassis to be clamped and used in two different orientations-horizontal and vertical. Still looking at FIG. **18**—ribs **126**, **128** are used to mount the rod and firearm support when it is clamped in the orientation shown. If the chassis **116** is rotated 90 degrees counter-clockwise and clamped in a vertical orientation, then ribs **118**, **120** are used to mount the rod and

firearm support. These options are explained in more detail in the following.

(51) FIG. 15 shows the alternate embodiment when clamped in a horizontal orientation to square tubing 64. In this context, “clamped in a horizontal orientation” means that screw clamps 44 exert clamping force in a horizontal orientation. The advancing screw clamps squeeze square tube 64 against clamping bracket 42—which preferably includes a vee-notch shape as shown so that it can be stably clamped to a square or round shape.

(52) Horizontal mount upper rib 126 includes a transverse horizontal mount upper hole 130. Horizontal mount lower rib 128 includes an aligned horizontal mount lower hole 132. These holes 130, 132 slidably receive smooth rod 90—which mounts firearm support 56 as for the prior versions. Smooth rod 90 also passes through hole 106 in lock plate 108. The far end of lock plate 108 is secured within a slot in unified chassis 116 (The slot is not visible in FIG. 15 but, but horizontal mount lock plate slot 136 is shown in the elevation view of FIG. 19).

(53) A unidirectional latch is thereby created. The free end of lock plate 108 is urged downward by gravity (supplemented by a spring if desired). If the user pulls up on firearm support 56, smooth rod 90 smoothly slides through hole 106 and any desired vertical extension for firearm support 56 can be easily set. Smooth rod 90 cannot move down, however, unless the user lifts up on the end of lock plate 108 (denoted as release 110). A compression spring may be added between lock plate 108 and upper rib 126 to pivot the lock plate downward. However, experimentation has shown that the spring is not essential. If the lock plate is made of a dense material—such as steel—then gravity alone is sufficient to rotate the free end downward and lock hole 106 against smooth rod 90.

(54) In this embodiment one can also easily remove the rod/support and the lock plate. These may be removed by pulling smooth rod 90 upward until it is free of all the holes 130, 106, 132 and removed entirely from unified chassis 116. This also allows lock plate 108 to be pulled to the right and free of horizontal mount lock plate slot 136 (shown in FIG. 19). The rod/support can then be placed in holes 122, 124 and the lock plate installed in a second corresponding slot.

(55) FIG. 16 shows this configuration. The reader will note that the unified chassis has been placed in a vertical orientation—meaning that screw clamps 44 now exert clamping force in a vertical orientation. This is a good orientation for clamping to a horizontal object such as shelf 131. The far end of lock plate 108 is placed within a second slot in the chassis (The slot is not shown in FIG. 16 but is shown as vertical mount lock plate slot 134 in FIG. 17). Hole 106 in the lock plate is aligned with vertical mount upper hole 122 and vertical mount lower hole 124. Smooth rod 90 is then passed downward through the three aligned holes 122, 106, 124 to again create a unidirectional locking mechanism.

(56) In reviewing the orientations shown in FIGS. 15 and 16, the reader will note how the features of unified chassis 116 allow the invention to be clamped in a wide variety of orientations while still providing a vertical adjustment range for smooth rod 90 and firearm support 56.

(57) FIG. 20 provides a section view through the middle of unified chassis 116 (note position of callouts in FIG. 19). This section view allows the reader to easily perceive the location of the aligned holes and the two lock plate slots. Vertical mount upper rib 118 includes vertical mount upper hole 122. Vertical mount lower rib 120 includes vertical mount lower hole 124. Vertical mount lock plate slot 134 is placed nearby as shown. Horizontal mount upper rib 126 includes horizontal mount upper hole 130. Horizontal mount lower rib 128 includes horizontal mount lock plate slot 136 as shown.

(58) FIG. 21 shows the same section location with the lock plate and the smooth rod shown installed. FIG. 21 shows both orientations in use at the same time. This is shown for convenience. In reality—only one of the two orientations would be in use at a time. For the horizontal clamping configuration, smooth rod 90 is placed through the aligned holes in ribs 126, 128. The reader will note how the inward end of lock plate 108 rests within horizontal mount lock plate slot 136, with smooth rod 90 passing through hole 106 in the lock plate. The rod may be pulled freely in the direction indicated by the arrow, but the lock plate will prevent motion in the opposite direction

(gravity will cause the lock plate to rotate clockwise until it rests in the latched position shown). (59) For the vertical clamping configuration, the smooth rod **90** is installed in ribs **118, 120**. In FIG. **21** the smooth rod is shown in the horizontal configuration but in actual use the assembly shown will be rotated approximately 90 degrees in the counterclockwise direction from the orientation shown in FIG. **21**. The reader will note how the inward end of lock plate **108** rests within vertical mount lock plate slot **134**. This again creates a unidirectional latch. Smooth rod **90** can be pulled freely in the direction shown by the arrow. However, it cannot move in the opposite direction since gravity will rotate lock plate **108** in the counterclockwise direction and cause hole **106** in lock plate to lock against smooth rod **90**.

(60) Each sets of ribs (**118, 120** or **126, 128**)—along with the aligned holes mounted on each rib—therefore comprise a new type of rod receiver that performs the same function as the more complex rod receiver shown in FIGS. **12-14**. The use of the unified chassis reduces the number of parts required to perform the same functions.

(61) The reader will thus perceive how the embodiment of FIGS. **15** and **16** allows the invention to be clamped in a horizontal orientation, a vertical orientation, and other orientations therebetween. In the version shown no stop is provided on the end of smooth rod **90**—so that the smooth rod can be pulled entirely free of unified chassis **116**. No stop is provided on lock plate **108** either. This allows the smooth rod and the lock plate to be removed and reinstalled in the other holes and slot in order to transition from the horizontal to the vertical configuration or vice versa. In other versions a stop and/or a spring can be provided. The stop can be made removable so that the smooth rod can still be transitioned between the vertical and horizontal orientations.

(62) Many other features and combinations of features can be provided for the invention. Examples include: 1. Aluminum can be used for chassis **34** and steel for the adjustable rod extending from the rod receiver. 2 The weldment shown for the chassis can be a unified piece made by various processes—such as extrusion. 3. The holes in the lock plates can be slots instead. 4. Many different types of screw clamps can be substituted for the type shown. An entirely different clamping mechanism can also be used—such as a caulking gun drive mechanism. 5. The firearm support can be any desired shape instead of the simple “vee” shown. 6. The firearm support can be covered with a compliant material such as a natural or synthetic rubber. 7. Many alternate designs for clamp bracket **42** can be substituted.

(63) The preceding descriptions contain significant detail regarding the novel aspects of the present invention. They should not be construed, however, as limiting the scope of the invention but rather as providing illustrations of the preferred embodiments of the invention. Many other embodiments will occur to those skilled in the art. Thus, the scope of the invention should be fixed by the following claims, rather than by the examples given.

Claims

1. An adjustable steady rest for a user aiming a firearm, comprising: (a) a chassis, including a clamp mount and a clamping bracket; (b) a clamp attached to said clamp mount and configured to clamp against said clamping bracket; (c) said chassis including a rod receiver with a first hole and a second hole; (d) a rod passing through said first and second holes, said rod being translatable with respect to said rod receiver, an amount of extension of said rod from said rod receiver being selectively adjustable by said user; (e) a firearm support mounted on a distal end of said rod; and (f) wherein said adjustable engagement between said rod and said rod receiver comprises a unidirectional latch configured to allow said user to pull said rod freely in a first direction but not in a second direction opposite said first direction.
2. The adjustable steady rest as recited in claim 1, wherein said rod receiver comprises: (a) a first rib, with said first hole passing through said first rib; and (b) a second rib offset from said first rib, with said second hole passing through said second rib and said second hole being aligned with said

first hole.

3. The adjustable steady rest as recited in claim 1, wherein said unidirectional latch comprises: (a) a lock plate having a first end, and a second end, and a lock plate hole; (b) said chassis having a lock plate slot; and (c) said first end of said lock plate resting within said lock plate slot and said rod passing through said lock plate hole.

4. The adjustable steady rest as recited in claim 2, further comprising: (a) a third rib having a third hole, said third rib being oriented transversely to said first rib; (b) a fourth rib offset from said third rib and having a fourth hole aligned with said third hole; and (c) said third and fourth holes being configured to receive said rod.

5. The adjustable steady rest as recited in claim 1, wherein said adjustable engagement between said rod and said rod receiver comprises: (a) said rod having a smooth, cylindrical exterior; (b) a lock plate, including a passage configured to bind against said cylindrical exterior of said rod when said lock plate is tilted; (c) a spring configured to bias said lock plate to said tilted position; and (d) a release configured to bias said lock plate away from said titled position.

6. The adjustable steady rest as recited in claim 1, wherein said clamp is a clamping screw.

7. The adjustable steady rest as recited in claim 6, wherein said clamping bracket is aligned with an axis of rotation of said clamping screw.

8. The adjustable steady rest as recited in claim 7, wherein said clamping bracket comprises: (a) a first leg angularly offset 45 degrees in the anti-clockwise direction from said axis of rotation of said clamping screw; and (b) a second leg angularly offset 45 degrees in the clockwise direction from said axis of rotation of said clamping screw.
